Intent Compression Architecture (ICA)

Clarification-First Ambiguity Management for Large Language Models

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Executive Summary

Large language models often answer the literal wording of a question even when the intended meaning depends on unstated assumptions or definitions. When multiple interpretations are plausible the model must hedge across them, producing broader responses that use more tokens and feel less precise.

Intent Compression Architecture (ICA) introduces a clarification-first reasoning step. If ambiguity could change the correct answer the system asks a short clarification question. Once the user confirms intent the final response is generated conditioned on that confirmed meaning.

Motivating Example

Example question: "Is Elon Musk spreading right-wing propaganda?" The word propaganda has multiple definitions. Some users mean persuasive advocacy while others mean coordinated deceptive messaging. ICA resolves the ambiguity by asking the user which definition they mean before generating the final answer.

System Architecture

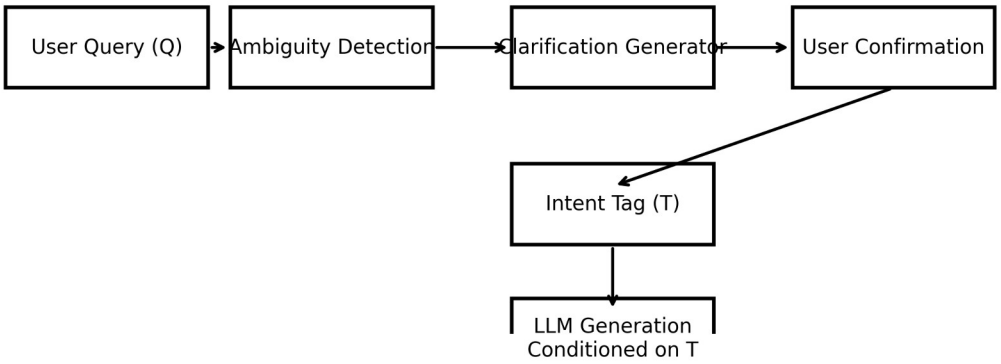


Figure 1. ICA architecture showing clarification-first interaction flow.

Ambiguity Taxonomy

Type	Description
Definition ambiguity	Terms whose meaning depends on interpretation such as propaganda or misinformation.
Constraint ambiguity	Missing parameters such as time range, location, runtime environment or dataset.
Referential ambiguity	Unclear references where multiple entities could match a description.
Causal ambiguity	Questions about causality requiring thresholds or counterfactual reasoning.
Task ambiguity	User intent may involve explanation, evaluation, summarisation or generation.

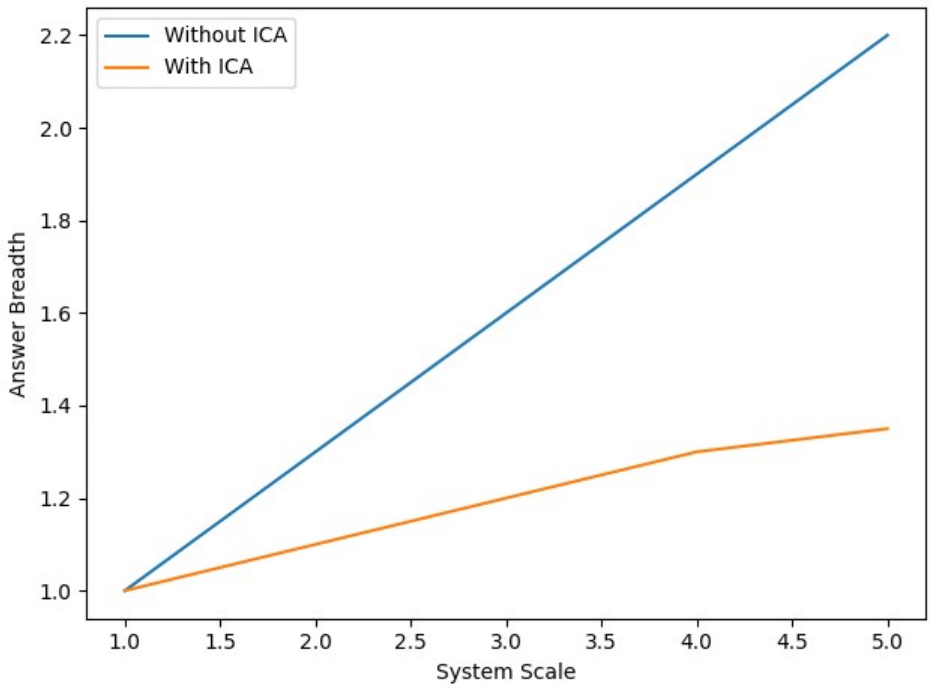
Interpretive Entropy Model

$$H(I \mid Q) = -\Sigma P(I_i \mid Q) \log P(I_i \mid Q)$$

When interpretive entropy is high the model must hedge across interpretations. ICA reduces entropy before answer generation by confirming intent.

Reverse Funnel Hypothesis

As AI systems scale interpretive variance increases. Different users attach different meanings to the same terms. This can cause answers to become broader over time.

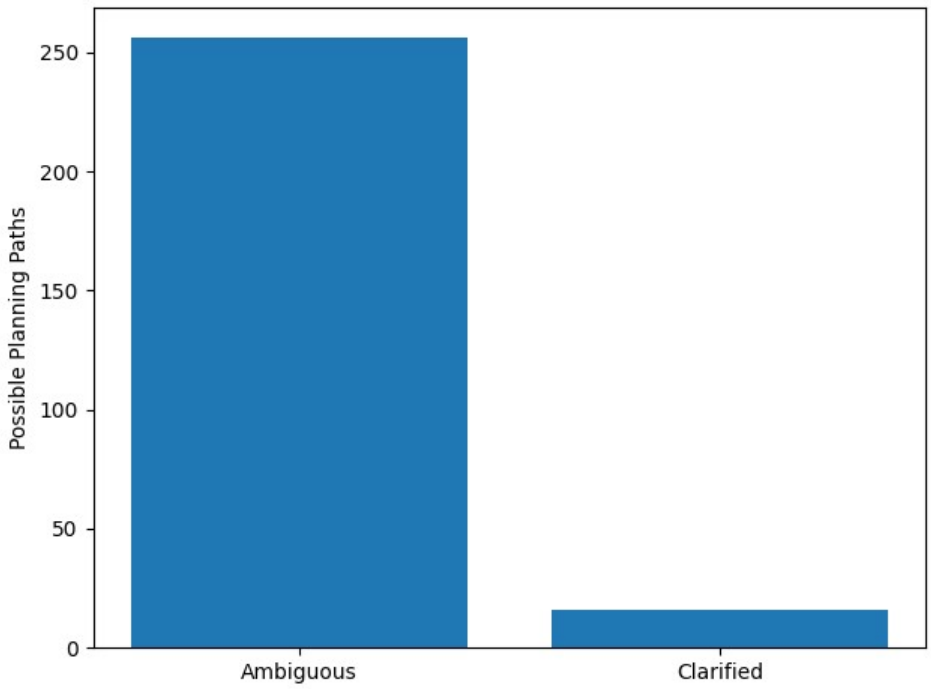


Token Economics

Example token comparison:
Hedged response: 280 tokens
Clarification: 22 tokens
Conditioned response: 140 tokens
Total ICA tokens: 162 (~42% reduction)

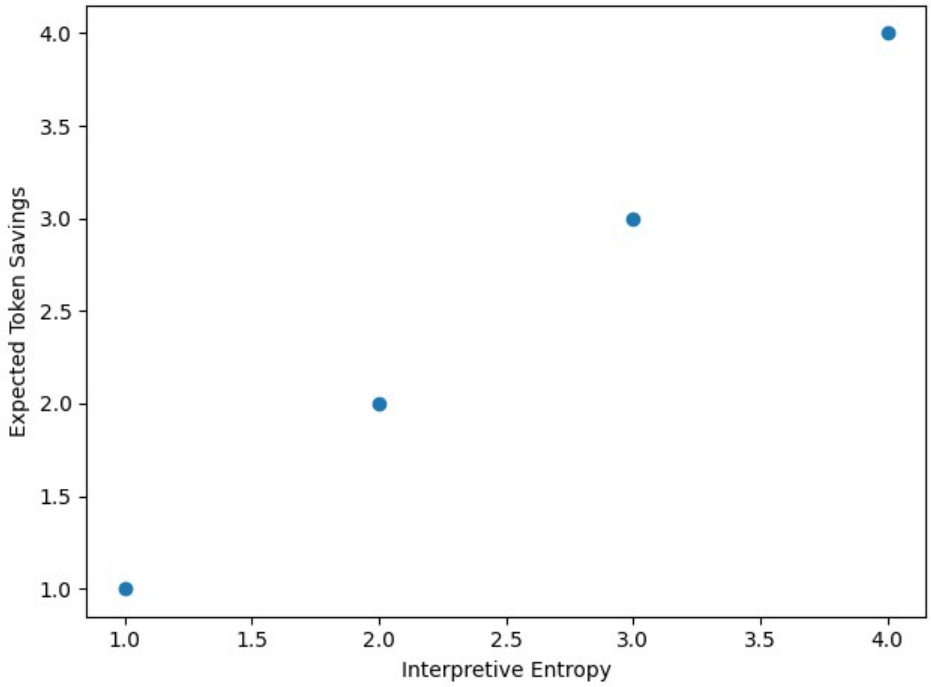
Agent Branching Model

In agent systems ambiguity increases the branching factor of planning trees. If b is branching factor and d is depth then complexity approximates O(b^d).



Clarification Decision Boundary

The system decides whether to clarify based on estimated interpretive entropy and expected token savings.



Evaluation Framework

Metric	Definition	Purpose
Total Tokens per Resolved Task	All tokens used until task completion	Measures compute cost
Clarification Frequency	How often clarification occurs	Measures user friction
Agent Loop Depth	Average retries in agent workflows	Measures reliability
Response Specificity	Reduction in hedging after clarification	Measures precision

Conclusion

Intent Compression Architecture proposes a structural approach to ambiguity management. By confirming meaning before answering, models can improve precision, reduce token usage, and stabilise agent workflows.

Visual Summary of ICA Concept

This diagram summarises the core idea behind Intent Compression Architecture. Instead of answering an ambiguous query directly, the system identifies possible interpretations, asks a minimal clarification question, confirms the user's intent, and then generates a precise answer conditioned on that confirmed meaning.

Figure: Visual summary of clarification-first interaction flow.